

Look-Ahead Bias Audit — One-Page Sign-Off

Project: Backtesting the ProntoNLP Earnings-Call ATC Signal · Author: Yueqi Lin · Reference: Student Handout §3

#	Item	Status & Evidence (notebook cell / test ID)
3.1	Entry timing (AMC vs BMO)	PASS — hour $\geq$ 16 UTC -> next NYSE day; hour $<$ 16 UTC -> same day. Assertion in 00_data_prep.ipynb cell 19. Gray-zone (13–15 UTC) rule documented in cell 18.
3.2	Forward returns are targets only	PASS — {return_1d ... return_20d} disjoint from feature_cols (cell 24). T2: 0 overlaps / 772 features.
3.3	Cross-sectional features are PIT	PASS — All Z-scores / sector ranks deferred to 01_analysis.ipynb walk-forward; per-fold on training events only.
3.4	Feature selection inside training	PASS — IC top-30 sparse selection re-fit on each fold's training data; no full-sample selection. T15 (07_audit_gap_tests.py) asserts selection is data-dependent (Jaccard 0.05 across disjoint windows).
3.5	Imputation / scaling inside training	PASS — StandardScaler.fit(train) then transform(test) inside walk-forward loop.
3.6	Universe membership is PIT	PASS — RU3K fully PIT via CRSP top-3000 mcap (annual June reconstitution; T11 = 0 violations / 153,988 events). SP500/SP1500 current-composition caveat documented per handout §6.3.
3.7	INGESTDATEUTC is not availability date	PASS — Inspected: 1,658d mean lag (batch backfill). Entry date uses MOSTIMPORTANTDATEUTC only; rationale in cell 18. T6: 0 INGEST features in model input.
3.8	No “future” QoQ deltas	PASS — All QoQ/2Q/YoY deltas use groupby(BESTTICKER).shift(1/2/4) on data sorted by entry_date ascending. T8 verified.
3.9	Corporate-action / delisting	PASS — NaN-return events excluded from all analyses (no roll-forward, no imputation). Documented in 00_data_prep.ipynb §9b and research PDF §7. T16 (07_audit_gap_tests.py) asserts no mass fillna(0) in return_20d.
3.10	Hyperparameter tuning	PASS — Tuning sub-period 2010–2017; walk-forward test 2018Q1+ uses frozen hyperparameters. No full-sample grid search. T17 (07_audit_gap_tests.py) asserts no GridSearchCV / Optuna / hyperopt in 01_analysis.ipynb.

Programmatic verification: T1–T8 in 02_lookahead_tests.ipynb, T9–T14 in 06_wrds_lookahead_tests.py, T15–T17 in 07_audit_gap_tests.py — all 17 tests PASS.

Detailed implementation notes for each item are preserved in reports/look_ahead_audit_detail.md.

I certify that all ten look-ahead audit items above have been reviewed and pass. The backtest implementation in 00_data_prep.ipynb and 01_analysis.ipynb contains no look-ahead bias that I am aware of.

Signed: Yueqi Lin · Date: May 2026 · Course: LLM-Driven Quant Research — Final Assignment